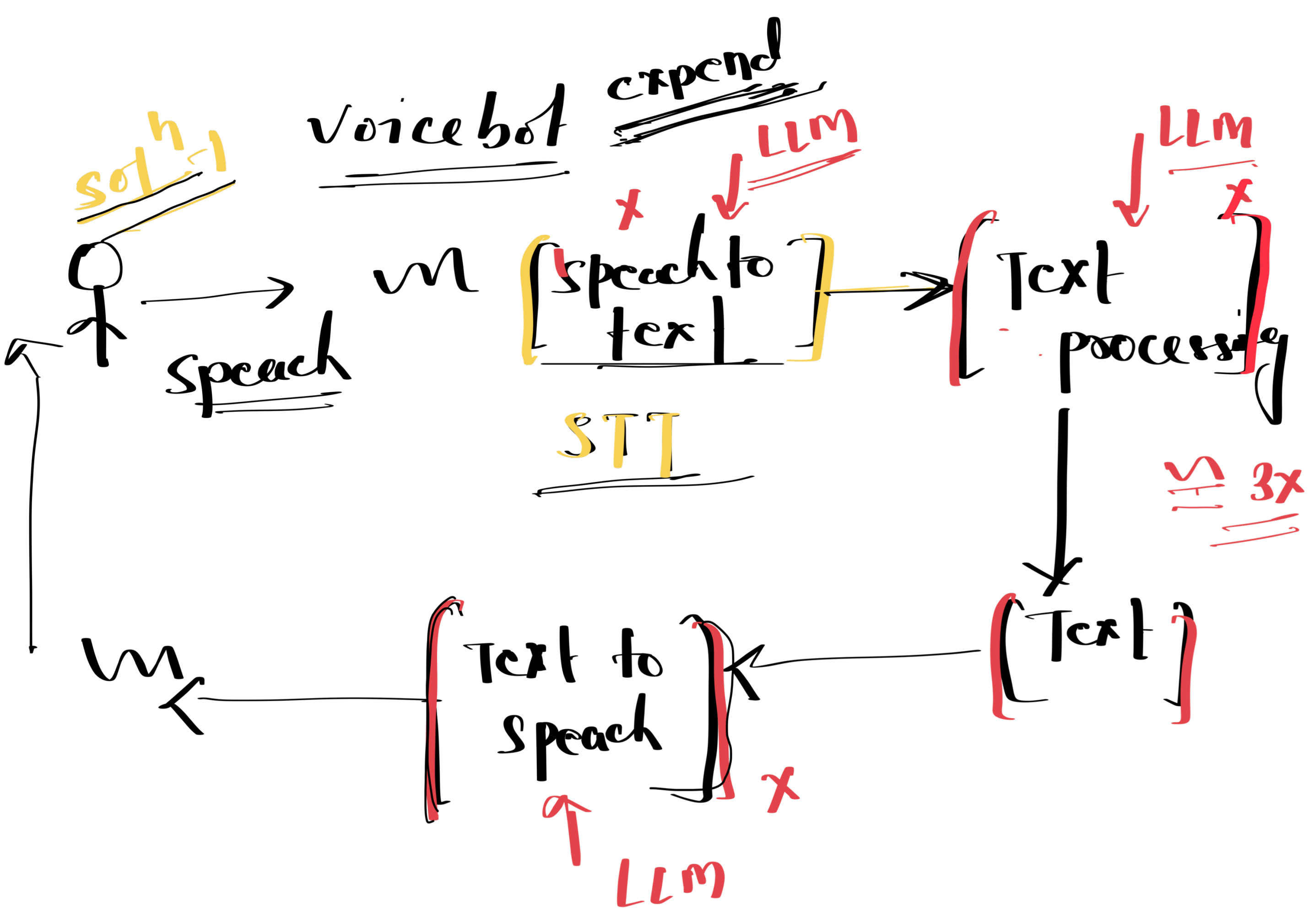
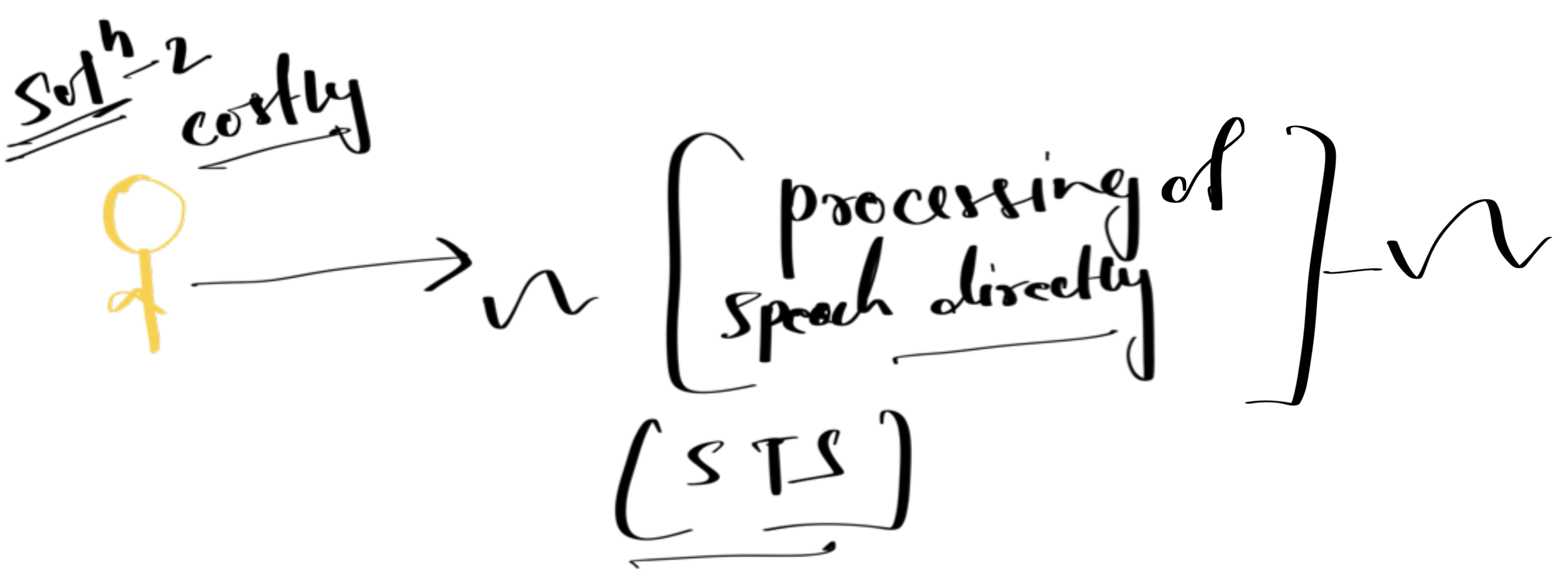
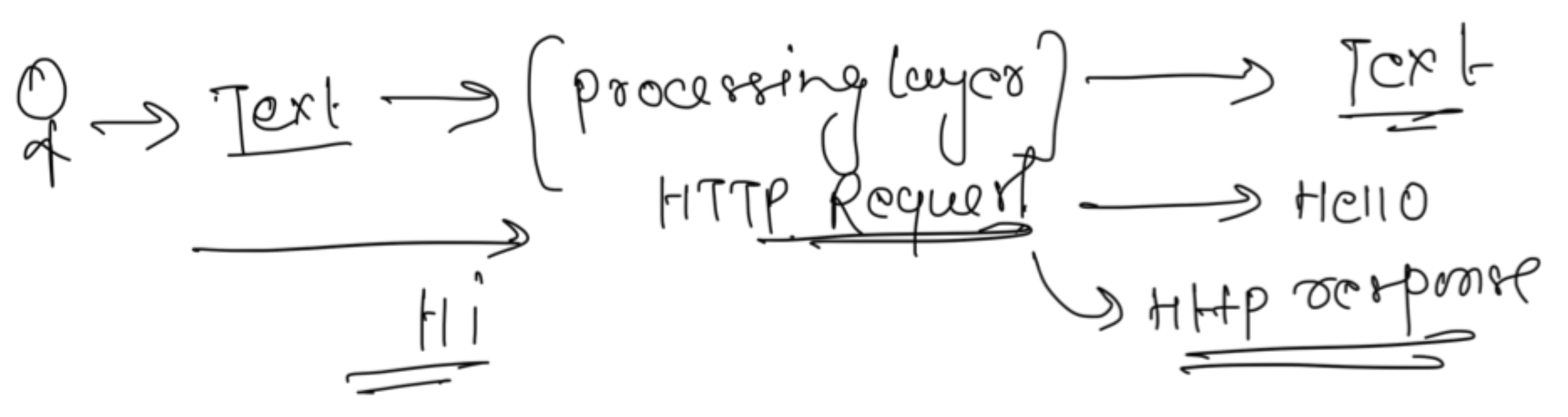


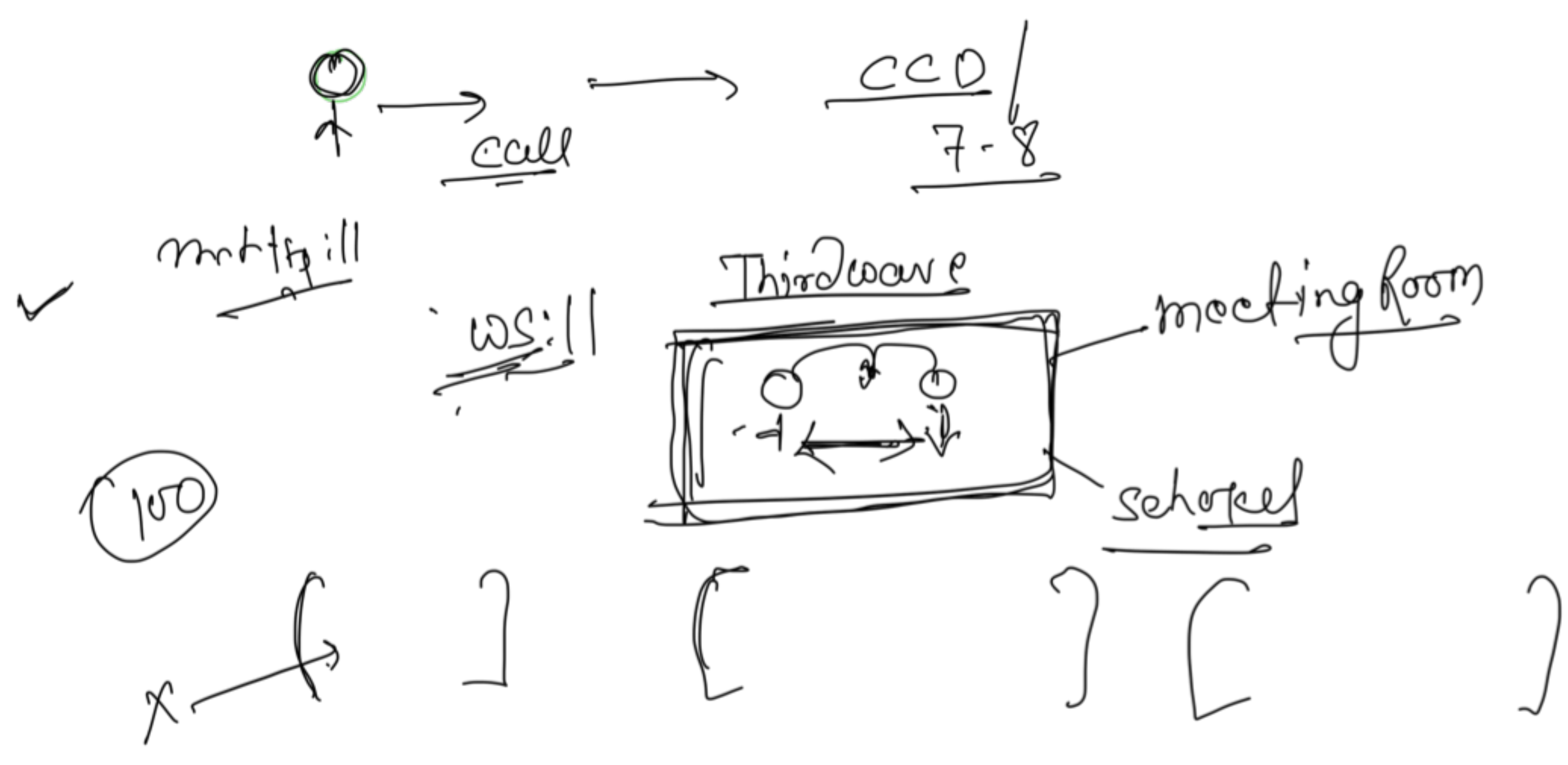


why voice bot building has heavy Engineering lift

conventional chatbot



WebSchoket



Livekit

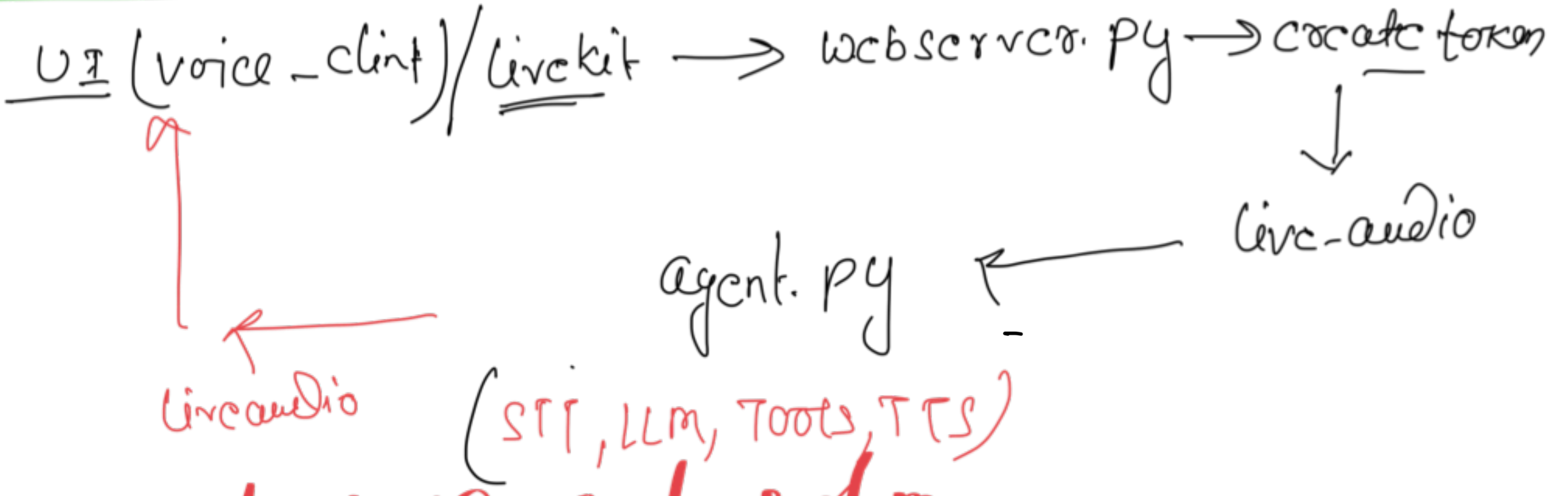
① STT
② TTS
③ Text processing engine
④ Live kit

idea

The core idea of voice bot

→ STT ( ) → Brain (LLM) + Hands (Tools) + mouth (TTS)

① STT (speech to text)	kokoro / speechmatics
② LLM	openAI
③ Tools	lookup_booking / update booking dates
④ TTS	fish Audio / Serval AI
⑤ Transport	Livekit (WEBRTC)



steps to build such system

① create the env

↳ python -m venv .venv

② activate the env

↳ source .venv/bin/activate

③ create requirements. and install

pip install -r requirements.txt

Hotel- Tools

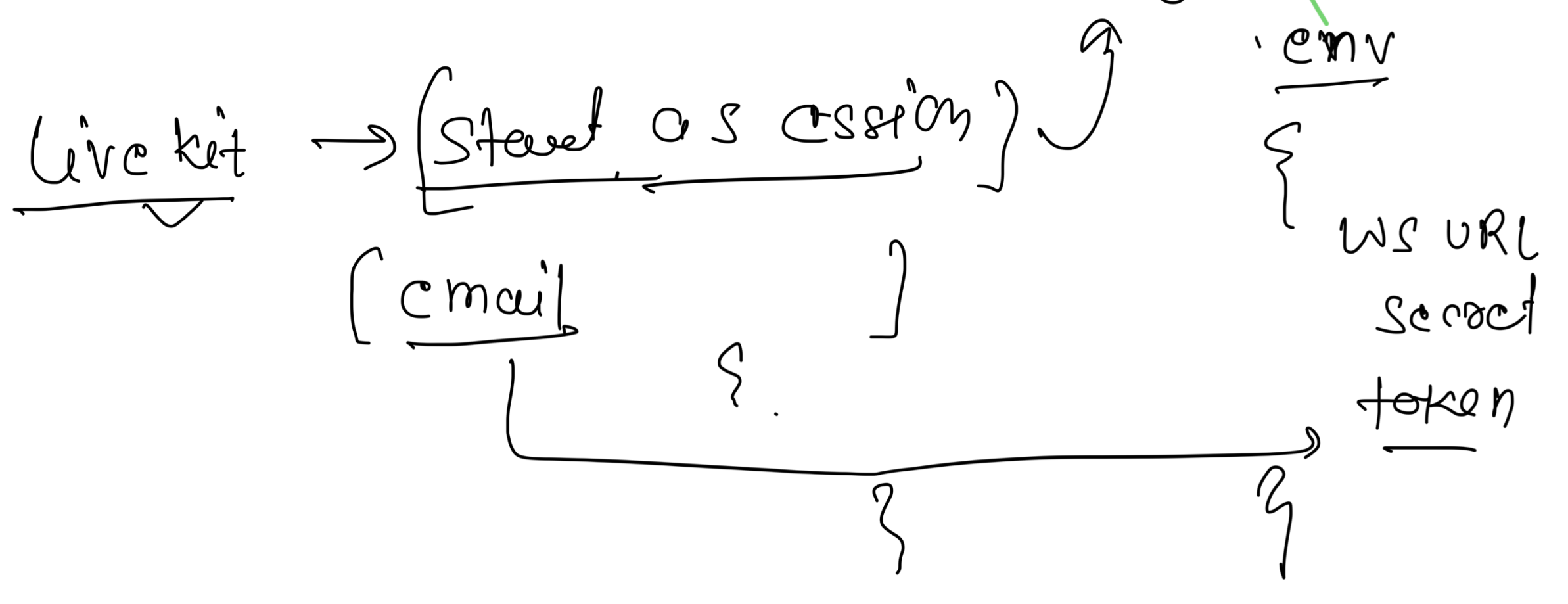
↳ lookup( )

↳ update( )

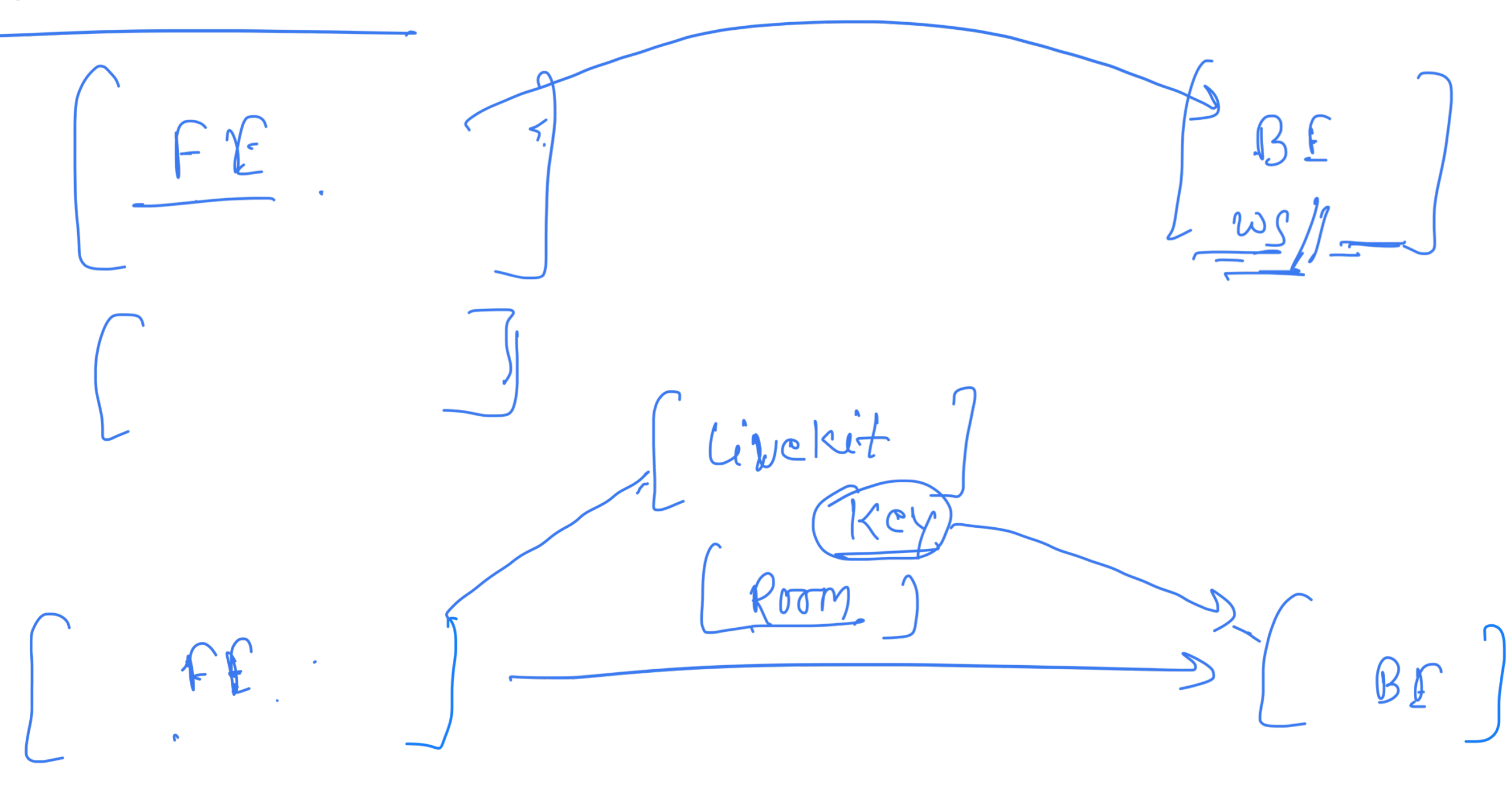
Building the custom livekit client

Run — agent.py → local

python agent.py dev



custom client



token_utils.py :-

→ The server creates a room and a token (a "ticket to join"), then call livekit to send the agent into the room.

→ I never give the browser your API key. Short-time key

Webserver.Py

it is fast API with 2 routes

① / serves the page → UI

② /api/session → return the token → build_token.py